

Integration of Rider Premiums and Benefits in Multi-Component Life Insurance Models

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Overview

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- Extension of an existing life insurance product by adding rider components
- Joint modeling of:
 - main risk (base contract)
 - supplementary benefits (riders)
 - investment component
- Requirement: consistency in premiums, benefits and reserve projection

Core Modeling Problem

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- Input payment must be decomposed into multiple components
- Previous model deducted only the main risk component, while the extended model consistently separates main risk and all rider components before allocating the remaining part to investment
- Each component has different actuarial behavior:
 - mortality-driven (main risk)
 - morbidity-driven (riders)
 - financial accumulation (investment)
- All components must be integrated into a single projection framework

Mathematical Foundations

- Premium decomposition across main risk, rider components, and investment part

$$\text{Entry} = P_{\text{main}} + \sum_i P_{\text{rider},i} + P_{\text{inv}}$$

- Present value decomposition:

$$V = \sum \text{PV}(\text{benefits}) - \sum \text{PV}(\text{premiums})$$

- Example structure:

$$A_{x:\overline{n}|}^1 \quad (\text{main risk}), \quad A_{x:\overline{n}|}^{1(r)} \quad (\text{rider})$$

- Annuity-type premium streams:

$$\ddot{a}_{x:\overline{n}|}$$

- Multiple biometric bases may be involved

Rider Structure

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- Each rider modeled as an independent component:
 - own tariff parameters
 - own probability structure
 - own benefit cash-flow
- Typical rider categories:
 - death / disability extensions
 - critical illness (dread disease)
 - dependent-related risks
- Aggregation performed at projection level

Combination of Risk Bases

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- Multiple insured entities:
 - main insured
 - partner / child (depending on rider)
- Different biometric tables may apply
- Resulting probabilities enter:

$${}_nE_x$$

- Combined impact reflected in:
 - premium calculation
 - expected benefits

Projection Framework

- Monthly projection of contract value
- Generic update step:

$$V_{t+1} = (V_t + P_{\text{inv}} - C_{\text{risk}}) \cdot (1 + r)$$

- Risk costs:

$$C_{\text{risk}} = P_{\text{main}} + \sum_i P_{\text{rider},i}$$

- All components interact within a single state variable V_t

Implementation Aspects

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- Modular structure:
 - premium calculation (riders)
 - benefit calculation
 - projection engine
- Configuration-driven activation of riders
- Parameterization via external inputs
- Integration into existing system without redesign

Engineering Challenges

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- Consistency between:
 - Excel specification
 - production implementation
- Backward compatibility with previous product version
- Handling heterogeneous rider definitions
- Ensuring numerical stability in long-term projections

What was actually solved

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- Separation of premium into independent components
- Consistent inclusion of riders in:
 - pricing
 - benefit modeling
 - reserve projection
- Unified projection of multi-component contract
- Preservation of compatibility with existing model structure

Outcome

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- Extended product supporting multiple supplementary benefits
- Transparent decomposition of cash-flows
- Reusable modular structure for future extensions
- Stable integration into existing actuarial framework